

Replication README File:

“The Geography of Child Penalties and Gender Norms: A Pseudo-Event Study Approach” by Henrik Kleven

1 Overview

The code for this project is provided as R programs and is contained in the folder `Code` of this replication package. These programs are used to clean raw data, create pseudo-panel data, and generate all figures and tables in the paper. The main program is `MASTER.R`. Running this program performs the full replication pipeline. Before running `MASTER.R`, the program `0_setup.R` must be run once to install all packages required for the replication pipeline. The program `setup.R`, which is sourced automatically when `MASTER.R` is run, loads these packages, defines file paths and parameters used throughout the code, and sources `functions.R`. The program `functions.R`, in turn, defines the main functions used throughout the project.

The data used in this project come from several different sources. With the exception of two data sets, all raw data are contained in the folder `Data/Raw_Data` of this replication package. Due to license restrictions, we are unable to distribute the PSID data through the folder `Data/Raw_Data` and cannot distribute the GSS geocode data at all. Code sections relying on PSID data or GSS geocode files are executed conditionally, depending on local data availability. If these data are not present, the corresponding sections are skipped automatically. We provide the raw PSID data via a PSID-compliant deposit. We also provide derived GSS datasets in the folder `Data/Working_Data` of this replication package, which allow all figures based on GSS data to be reproduced without access to the raw GSS geocodes.

2 Data Availability and Provenance Statements

Statement About Rights

The author of the manuscript has legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

Most of the data used for this project are produced by the US government and are in the public domain. In compliance with data publishing rights, all raw data are provided in the folder `Data/Raw_Data` of this replication package, with the exception of the PSID data and the GSS geocode data. We provide the raw PSID data via a PSID-compliant deposit. We cannot distribute the GSS geocode data at all, but we do provide derived GSS datasets in the folder `Data/Working_Data` of this replication package.

Details on Each Data Source

Below, we list each data source and its availability in detail:

1. **Current Population Survey (CPS) Data:** We combine the Basic Monthly Files for 1989-2020 and the Annual Social and Economic Supplement (ASEC) for 1968-2020. The sample used in the matching procedure is composed of:

- Childless individuals aged between 20 and 45 in the years 1968-2020
- Parents with age at first birth between 25 and 45 in the years 1973-2020, interviewed between event times 0 and 10

We provide these data in the file `raw_cps.RData`, located in the folder `Data/Raw_Data/CPS` of this replication package. CPS data are also publicly available and can be downloaded from [here](#).

2. **American Community Survey (ACS) Data:** We use ACS data between 2000 and 2019. The sample used in the matching procedure is composed of:

- Childless individuals aged between 20 and 45
- Parents with age at first birth between 25 and 45, interviewed between event times 0 and 10

We provide these data in the file `raw_acs.RData`, located in the folder `Data/Raw_Data/ACS` of this replication package. ACS data are also publicly available and can be downloaded from [here](#).

3. **National Longitudinal Survey of Youth (NLSY) – 1979 Cohort Data:** We use NLSY79 data between 1979 and 2018. These data are provided in the file `raw_nlsy.RData`, located in the folder `Data/Raw_Data/NLSY79` of this replication package. These data are also publicly available [here](#). To create the corresponding data extract and accompanying documentation on this website, users need to upload a tagset under the “Tagset” tab. A tagset is a file containing a list of selected variables. The variables used in this project are contained in the tagset `NLSY79.NLSY79`, which we provide in the `Data/Raw_Data/NLSY79` folder.

4. **Panel Study of Income Dynamics (PSID) Data:** We use a combination of variables from the Family-level and Individual-level PSID data, as well as the Childbirth and Adoption History data provided in the Family History files. We use PSID data between 1968 and 2019. We provide these data through a PSID-compliant deposit, available at the DOI <https://doi.org/10.3886/ICPSR301089.V1>.
5. **General Social Survey (GSS) Data:** We merge raw GSS data with geocode data, which contain state identifiers, using the variables `id` and `year`. We use these data to construct the Gender Progressivity Index (GPI). To measure gender progressivity consistently over time, we focus on three questions available in all five decades of the data. These questions ask respondents whether they strongly agree, agree, disagree, or strongly disagree with the following statements:
 - It is much better for everyone involved if the man is the achiever outside the home and the woman takes care of the home and family
 - A working mother can establish just as warm and secure a relationship with her children as a mother who does not work
 - A pre-school child is likely to suffer if his or her mother works

We construct the GPI based on the average standardized response to these questions. We provide the raw GSS data in the file `GSS7218_R1.dta`, located in the folder `Data/Raw_Data/GSS` of this replication package. Access to the geocode data must be requested separately and involves a cost of several thousand dollars. For this reason, we also provide derived data containing the measures of gender progressivity by state and `state×year`, which are used to produce Figures 9-11 and Figure A.17. These derived data can be found in the files `gss_imputed_state.rds` and `gss_imputed_state_by_decade.rds`, located in the `Data/Working_Data` folder.
6. **2019 State Population Data from the US Census Bureau:** For some specifications, we require population weights by state. Using 2019 state population data from the US Census Bureau, we calculate the share of the 2019 US population living in each state. These data are included in the file `nst-est2019-alldata.csv`, located in the `Data/Raw_Data/Census Bureau Population Data` folder of this replication package. The data are also publicly available [here](#).
7. **Global Child Penalties:** We use the publicly available global child penalties dataset from [The Child Penalty Atlas](#) by [Kleven, Landais, and Leite-Mariante \(2025\)](#). We provide these data in the file `global_penalties.xlsx`, located in the folder `Data/Raw_Data/Global Child Penalties` of this replication package. The replication package for [Kleven, Landais, and Leite-Mariante \(2025\)](#) is available at [10.5281/zenodo.13742225](https://zenodo.org/record/13742225).

3 List of Raw Datasets

The following table lists the raw datasets described in the previous section and indicates whether and where they are provided in this replication package.

Table 1: Raw Data Sources

Data Name	Data File	Location	Citation
Current Population Survey (CPS) Data	<code>raw_cps.RData</code>	<code>Data/Raw_Data/CPS</code>	CPS (1968-2020)
American Community Survey (ACS) Data	<code>raw_acs.RData</code>	<code>Data/Raw_Data/ACS</code>	ACS (2000-2019)
National Longitudinal Survey of Youth (NLSY) – 1979 Cohort Data	<code>raw_nlsy.RData</code>	<code>Data/Raw_Data/NLSY79</code>	NLSY (1979-2018)
Panel Study of Income Dynamics (PSID)	Provided through deposit 10.3886/ICPSR301089.V1		PSID (1968-2019)
General Social Survey (GSS)			GSS (1972-2018)
Raw GSS Data	<code>GSS7218.R1.dta</code>	<code>Data/Raw_Data/GSS</code>	
Geocode Data	Not provided (access must be requested separately)		
2019 State Population Data from the US Census Bureau	<code>nst-est2019-alldata.csv</code>	<code>Data/Raw_Data/Census Bureau Population Data</code>	SPD (2019)
Global Child Penalties	<code>global_penalties.xlsx</code>	<code>Data/Raw_Data/Global Child Penalties</code>	Kleven, Landais, and Leite-Mariante (2025)

4 Computational Requirements

Software Requirements

The last validated run was performed on April 06, 2026, on a server, with a total run time of 63 hours. The pipeline is designed for server/HPC execution. First, the program `0_setup.R` must be run once to install all packages required for the replication pipeline. Then, the main program, `MASTER.R`, must be run. The following setup reflects the environment used for the most recent validation:

- **OS / Platform:** Windows Server 2019 x64 (build 17763); x86_64, mingw32
- **R:** R 4.5.2 (2025-10-31 ucrt); executed from Windows Terminal via `Rscript`; session timezone PDT
- **CPU/RAM/GPU:** Intel Xeon Gold 6146 @ 3.20GHz; 1.5 TB RAM; NVIDIA Quadro P2000 (5GB)
- **Master program:** `MASTER.R` (cluster/server execution recommended)

Controlled Randomness

The code uses the following seeds:

- `setup.R`, line 105: `seed.val <- 9746290`
- `table_validation_assumption_conditional_independence.R`, line 195: `set.seed(72839251)`

Memory, Runtime, and Execution

- **Memory:** ≈ 1.5 TB RAM node (end-to-end without swapping)
- **Runtime (indicative, based on the hardware described above):** ≈ 21 hours to build datasets from raw sources and ≈ 42 hours to run analyses
- **Execution:** Use a cluster/server to run `MASTER.R`. Local machines are not recommended

R Packages

All core R packages used in this project were installed from the Comprehensive R Archive Network (CRAN) at the versions specified below. The following list summarizes these packages:

- **Estimation:** `fixest` 0.12.1; `plm` 2.6.6; `sandwich` 3.1.1; `lmtest` 0.9.40; `survey` 4.4.2; `multcomp` 1.4.28
- **Matching/Validation:** `Matching` 4.10.15; `MatchIt` 4.7.2; `easyPSID` 0.1.2
- **Data IO/Preparation:** `data.table` 1.17.8; `ipumsr` 0.9.0; `haven` 2.5.5; `readr` 2.1.6; `readxl` 1.4.5; `here` 1.0.1
- **Graphics/Maps:** `ggplot2` 3.5.2; `cowplot` 1.2.0; `directlabels` 2025.6.24; `usmap` 0.8.0; `sf` 1.0.23; `rnaturalearth` 1.1.0
- **Tables/Output:** `kableExtra` 1.4.0; `stargazer` 5.2.3; `xtable` 1.8.4; `DT` 0.33
- **Miscellaneous:** `binsreg` 1.1; `ROCR` 1.0.11; `ggthemes` 5.1.0; `stringr` 1.6.0; `dplyr` 1.1.4; `tidyr` 1.3.1; `tidyverse` 2.0.0

The following table lists all packages in the validated R session, including base and recommended packages bundled with R for completeness.

Table 2: R Packages and Versions (Session: 2026-04-06)

Package	Version	Package	Version	Package	Version
abind	1.4.8	haven	2.5.5	rappdirs	0.3.3
AER	1.2.15	here	1.0.1	Rdpack	2.6.4
backports	1.5.0	hms	1.1.4	readr	2.1.6
BBmisc	1.13	htmltools	0.5.9	readxl	1.4.5
bdsmatrix	1.3.7	htmlwidgets	1.6.4	rlang	1.2.0
binsreg	1.1	httr	1.4.7	rmarkdown	2.30
boot	1.3.31	ipumsr	0.9.0	rnaturalearth	1.1.0
broom	1.0.9	kableExtra	1.4.0	rnaturalearthdata	1.0.0
cachem	1.1.0	KernSmooth	2.23.26	ROCR	1.0.11
car	3.1.3	knitr	1.51	rprojroot	2.1.1
carData	3.0.5	labelled	2.14.1	RStata	1.1.2
cellranger	1.1.0	LaF	0.8.6	rstudioapi	0.17.1
checkmate	2.3.3	lattice	0.22.7	sandwich	3.1.1
class	7.3.23	lifecycle	1.0.5	scales	1.4.0
classInt	0.4.11	lmtest	0.9.40	sessioninfo	1.2.3
cli	3.6.5	lubridate	1.9.4	sf	1.0.23
codetools	0.2.20	magrittr	2.0.4	SparseM	1.84.2
collapse	2.1.5	maps	3.4.3	stargazer	5.2.3
conflicted	1.2.0	MASS	7.3.65	stringi	1.8.7
cowplot	1.2.0	Matching	4.10.15	stringmagic	1.2.0
crayon	1.5.3	MatchIt	4.7.2	stringr	1.6.0
data.table	1.17.8	Matrix	1.7.4	survey	4.4.2
DBI	1.2.3	MatrixModels	0.5.4	survival	3.8.3
digest	0.6.39	matrixStats	1.5.0	svglite	2.2.2
directlabels	2025.6.24	maxLik	1.5.2.1	systemfonts	1.3.1
dplyr	1.1.4	memoise	2.0.1	textshaping	1.0.4
dreamerr	1.5.0	miscTools	0.6.28	TH.data	1.1.5
DT	0.33	mitools	2.4	tibble	3.3.0
e1071	1.7.16	mlr	2.19.2	tictoc	1.2.1
easyPSID	0.1.2	multcomp	1.4.28	tidyr	1.3.1
evaluate	1.0.5	mvtnorm	1.3.3	tidyselect	1.2.1
farver	2.1.2	nlme	3.1.168	tidyverse	2.0.0
fastmap	1.2.0	numDeriv	2016.8.1.1	tigris	2.2.1
fastmatch	1.1.6	parallelMap	1.5.1	timechange	0.3.0
fixest	0.12.1	ParamHelpers	1.14.2	tzdb	0.5.0
forcats	1.0.1	pillar	1.11.1	units	1.0.0
foreign	0.8.90	pkgconfig	2.0.3	usmap	0.8.0
Formula	1.2.5	plm	2.6.6	usmapdata	0.6.0
generics	0.1.4	plyr	1.8.9	uuid	1.2.1
ggplot2	3.5.2	proxy	0.4.28	vctrs	0.7.2
ggtext	0.1.2	purrr	1.1.0	viridisLite	0.4.2
ggthemes	5.1.0	quadprog	1.5.8	withr	3.0.2
glue	1.8.0	quantreg	6.1	xfun	0.54
gridExtra	2.3	R6	2.6.1	xml2	1.5.1
gridGraphics	0.5.1	RColorBrewer	1.1.3	xtable	1.8.4
gridtext	0.1.5	Rcpp	1.0.0	zeallot	0.2.0
gtable	0.3.6	rbibutils	2.4	zoo	1.8.14

5 Description of Programs

- `MASTER.R` is the main program and orchestrates the full replication pipeline
- `0_setup.R` installs all packages required to run `MASTER.R` and fixes package versions¹
- `setup.R` loads packages, defines file paths and parameters used throughout the code, and sources `functions.R`
- `functions.R` defines the main functions used throughout the project
- Programs sourced in the **Label and Create Raw Data Files** section of `MASTER.R` convert raw data files into R format
- Programs sourced in the **Data Cleaning** section of `MASTER.R` clean the raw datasets
- `matching.R` uses cross-sectional CPS and ACS datasets to create the pseudo-panel dataset used in the project's CPS/ACS-based analyses
- Programs sourced in the **Results** section of `MASTER.R` generate all exhibits shown in the paper

6 Instructions to Replicators

- Set the default path to the replication folder before running the code
- Verify that the folder structure matches the paths specified in the program `setup.R`
- Run the program `0_setup.R` once to install all packages required for the replication pipeline
- After package installation is complete, run the main program `MASTER.R`
- Each program begins by clearing objects from the workspace, preserving the log objects where applicable, and then sourcing `setup.R`

7 List of Outputs and Programs

The provided code reproduces:

- ☑ Hausman test statistics reported in the text of the paper
- ☑ All tables and figures presented in the paper

The following table lists the programs and output files corresponding to each exhibit in the paper, as well as to the Hausman test statistics reported in the text of the paper.

¹In clean Linux environments, installation of some R package dependencies may require additional system libraries, including PostgreSQL development headers (`libpq-dev`) and OpenMPI development headers (`libopenmpi-dev`).

Table 3: Output

Figure/Table #	Program	Output file	Note
Table 1	table_descriptive_stats_xsection.R	table_descriptive_stats_xsection.tex	
Table 2	table_descriptive_stats_pseudo_panel.R	table_descriptive_stats_pseudo_panel.tex	
Table A.1	table_validation_assumption_conditional_independence.R	table_assumption3_test.tex	
Table 3	table_descriptive_stats_movers_immigrants.R	table_descriptive_stats_movers_vs_immigrants.tex	
Figure 1	fig_validation_eventstudy.R	validation_pseudo_*, validation_actual_*	Multiple files
Figure 2	matching_panel.R	validation_pseudo_psid*	Multiple files
Hausman Test Statistics	hausman_test.R	Hausman test statistics	
Figure 3	fig_validation_within_panel_scatter.R	validation_scatter*	Multiple files
Figures A.1-A.3	matching_appendix.R	matching_spec*	Multiple files
Figure A.4	fig_validation_histogram.R	fertility_prediction.pdf	
Figure A.5	fig_validation_age1b.R	heterogeneity_age1b*	Multiple files
Figures 4, A.6-A.8	fig_time.R	child_penalties_timeseries*	Multiple files
Figure A.9	fig_time_unscaled.R	*_timeseries_unadjusted	Multiple files
Figure A.10	fig_time_pct_exp.R	fraction_explained_timeseries.pdf	
Figures 5, A.11-A.13	fig_state_eventstudy_heatmaps.R	child_penalties_state*	Multiple files
Figure 6	fig_state_eventstudy_heatmaps.R	child_penalties_state_heatmap*	Multiple files
Figure A.14	fig_state_scatterplots_gender_gaps.R	child_penalties_vs_gender_gaps*	Multiple files
Figure 7	fig_state_scatterplots_unadjusted.R	baseline_effects_*	Multiple files
Figures A.15-A.16	fig_shrinkage.R	shrinkage_*	Multiple files
Figure 8	fig_eventstudy_demographics.R	child_penalties_educ*, child_penalties_marital*, child_penalties_race*	Multiple files
Figure 9	fig_gss_heatmaps_state.R	gss_heatmap.pdf	
Figure 10	fig_gss_vs_penalties_timeseries.R	gss_timeseries.pdf	
Figure A.17	fig_gss_timeseries_by_state.R	gss_timeseries_by_state.pdf	
Figure 11	fig_gss_vs_penalties_scatterplots_state_time.R	gss_state_by_time_*	Multiple files
Figures 12, A.18-A.19	fig_epidemiological_us_eventstudy_movers_vs_stayers.R	epidemiological_movers_vs_stayers_*	Multiple files
Figures 13, A.20	fig_epidemiological_us_scatterplots.R	epidemiological_movers_vs_stayers_rawscatter_*	Multiple files

Table 3: Output (*continued*)

Figure/Table #	Program	Output file	Note
Figure 14, Figure A.21	fig.epidemiological_ foreign.eventstudy_ birth.country.R	epidemiological_ immigrants_*	Multiple files
Figure 15	fig.epidemiological_ foreign.eventstudy_deciles.R	epidemiological_ immigrants_*	Multiple files
Figure 16, Figure A.22	fig.epidemiological_ foreign.scatterplots.R	epidemiological_immigrants_ rawscatter_*.pdf, epidemiological_ immigrants_ binscatter_*.pdf	Multiple files
Figure A.23	fig.epidemiological_ foreign.eventstudy_education.R	epidemiological_ immigrants.child_ penalties.educ*	Multiple files
Figure A.24	fig.epidemiological_ foreign.eventstudy_cultural.R	epidemiological_ immigrants_assimilation*	Multiple files

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